Brendan Boyd

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biboyd.github.io

Education

SUNY Stony Brook University 2020-Present
Physics & Astronomy Department
PhD Candidate in Physics,
with Concentration in Astronomy

Michigan State University 2016-2020
College of Natural Science
Honors College
Bachelor of Science, Astrophysics
Minors in Math and CMSE

Publications

Refereed

Ferran Poca-Amorós et al. *On the Importance of the Convective Urca Process in 3D Simulations of a Simmering White Dwarf.* Accepted by The Astrophysical Journal. 2026. arXiv: `2602.21440 [astro-ph.SR]`. URL: `https://arxiv.org/abs/2602.21440`

Brendan Boyd et al. "3D Convective Urca Process in a Simmering White Dwarf". In: *The Astrophysical Journal* 979.2 (Jan. 2025), p. 216. DOI: `10.3847/1538-4357/ad9bb0`. URL: `https://dx.doi.org/10.3847/1538-4357/ad9bb0`

Alexander I. Smith et al. "pynucastro: A Python Library for Nuclear Astrophysics". In: *The Astrophysical Journal* 947.2, 65 (Apr. 2023), p. 65. DOI: `10.3847/1538-4357/acbaff`

Conference Proceedings

B. Boyd et al. "Approximating Convective Urca Cooling in a Simmering White Dwarf". In: *Journal of Physics Conference Series.* Vol. 2997. Journal of Physics Conference Series. IOP, Apr. 2025, 012006, p. 012006. DOI: `10.1088/1742-6596/2997/1/012006`

B. Boyd et al. "Sensitivity of 3D Convective Urca Simulations to Changes in Urca Reactions". In: *Journal of Physics Conference Series.* Vol. 2742. Journal of Physics Conference Series. IOP, Apr. 2024, 012001, p. 012001. DOI: `10.1088/1742-6596/2742/1/012001`. arXiv: `2311.07743 [astro-ph.SR]`

Zhi Chen et al. "A Framework for Exploring Nuclear Physics Sensitivity in Numerical Simulations". In: *Journal of Physics Conference Series.* Vol. 2742. Journal of Physics Conference Series. IOP, Apr. 2024, 012021, p. 012021. DOI: `10.1088/1742-6596/2742/1/012021`

Alexander Smith Clark et al. "pynucastro 2.1: an update on the development of a python library for nuclear astrophysics". In: *Journal of Physics Conference Series.* Vol. 2742. Journal of Physics Conference Series. IOP, Apr. 2024, 012003, p. 012003. DOI: `10.1088/1742-6596/2742/1/012003`

Software

Ann Almgren et al. *AMReX-Astro/MAESTROeX: MAESTROeX 25.02*. Version 25.02. Feb. 2025. DOI: 10.5281/zenodo.14811053. URL: https://doi.org/10.5281/zenodo.14811053

AMReX-Astro initial_models team et al. *AMReX-Astro/initial_models: Release 24.03*. Version 24.03. Mar. 2024. DOI: 10.5281/zenodo.10827428. URL: https://doi.org/10.5281/zenodo.10827428

Brendan Boyd et al. "SALSA: A Python Package for Constructing Synthetic Quasar Absorption Line Catalogs from Astrophysical Hydrodynamic Simulations". In: *The Journal of Open Source Software* 5.52, 2581 (Aug. 2020), p. 2581. DOI: 10.21105/joss.02581

Research Experience

Type Ia Supernovae Progenitor Modeling 2022-Present
Using the MAESTROeX hydrodynamic code to study the Convective Urca Process during simmering phase of white dwarfs.

Galactic Modeling Research 2019-2020
Studied ENZO simulations produced by FOGGIE collaboration to better understand the Circumgalactic Medium (CGM). Generated synthetic spectra of the CGM to better inform/compare to observations.

MSU Campus Observatory 2018-2019
Assisted in data collection and reduction using 24-inch telescope. Observed cataclysmic variables, supernovae and transiting exoplanets.

HAWC Research Group 2016-2018
Studied gamma ray sources using the HAWC Observatory. Mainly worked on improving the detection sensitivity through machine learning techniques.

Teaching Experience

Teaching Assistant - Stony Brook University

AST 248: "The Search for Life in the Universe". A course designed to give an overview of the current knowledge of life outside of Earth and how we are searching for it. Topics such as habitability in our solar system, biosignatures, Fermi Paradox, etc.

Fall 2021, Spring 2021, Fall 2020

Undergraduate Learning Assistant - Michigan State University

AST 208: "Planets & Telescopes", **AST 207:** "The Science of Astronomy", **ISP 205:** "Visions of the Universe".

Spring 2020, Fall 2019, Spring 2019

Computational Skills

Programming Languages:
Proficient in Python
Proficient in C++
Competent in MPI parallelism
Competent in OpenMP threading
Competent in FORTRAN

Invited Talks

FSU Astronomy Seminar 2024
Presented work on hydrodynamic simulations of the Convective Urca Process at the Florida State University astronomy seminar.

SNEx Group 2024
Presented work on hydrodynamic simulations of the Convective Urca Process at the SNEx group; a group focused on type Ia SNe research, both observational and theoretical.

SBU IACS Student Seminar 2023
Presented work at the Stony Brook IACS student seminar on the challenges associated with using numerical fluid simulations to study astrophysics, with particular focus on Type Ia Supernovae.

Conferences Attended

Nuclei in the Cosmos 2025 2025
Presented poster on new nuclear networks used in simulations of the Convective Urca Process at NIC - Nuclear Astrophysics conference.

American Astronomical Society Meeting 245 2025
Presented poster on recently published paper pertaining to the Convective Urca Process.

ASTRONUM 2024 2024
Presented a talk on simulations of the Convective Urca Process at ASTRONUM - International Conference on Numerical Modeling of Space Plasma Flows.

Nuclei in the Cosmos 2023 2023
Presented poster on simulations of the Convective Urca Process at NIC - Nuclear Astrophysics conference.

ASTRONUM 2023 2023
Presented a poster on simulations of the Convective Urca Process at ASTRONUM - International Conference on Numerical Modeling of Space Plasma Flows.

American Astronomical Society Meeting 241 2023
Presented poster on simulation of the Convective Urca Process at the winter AAS Meeting

Summer Schools Attended

International High Performance Computing Summer School 2022
Summer school for early-career computational scientists. Familiarized students with major state-of-the-art aspects of HPC and Big Data Analytics. Provided advanced mentoring and facilitated international networking.

Certification

CIMER Mentoring Training 2024
Completed a training workshop grounded in the Entering Mentoring series from the Center for the Improvement of Mentored Experiences in Research

Community Service and Outreach

MSU Observatory Public Nights 2018

Open house events at the MSU Observatory. Taught the public about astronomy and the work done at the observatory. Assisted people with looking through telescopes.

MSU Science Festival 2017-2018

Annual event used to inform and inspire the general public. Worked a station explaining the HAWC experiment and Cosmic Rays.

Tour de Ville 2016-2020

Annual charity bicycle ride put on by the Northville Rotary Club. Help with sending mass emails for the event as well as contributing the day of the ride (e.g. setting up of rest stops along course)

Honors and Awards

Peter B. Kahn Prize 2024

Awarded to Physics & Astronomy graduate student for outstanding research and travel.

Hantel Fellowship 2017, 2019

Awarded to Michigan State undergraduates conducting research in physics.

Michigan State Dean's List 2016-2020

Recognized for eight semesters as a student with at least a 3.5 GPA.